



Data Analysis & Business Understanding

Week 1 Task

Submission Deadline

Saturday, 4 July 2026

Objective of the Task

A retail company wants to understand customer purchase behavior before investing in a recommendation system. Acting as a data analyst, you are required to explore the provided dataset, identify patterns and data quality issues, and produce a complete Exploratory Data Analysis (EDA) report that helps the company understand who its customers are, which products sell the most, and which countries generate the most revenue.

Dataset

- **Name:** Online Retail II Dataset
- **Link:** <https://archive.ics.uci.edu/ml/datasets/Online+Retail+II>
- **Description:** Real transaction data from a UK-based online retailer (2009–2011), including invoices, product descriptions, quantities, prices, and customer IDs.

Requirements

- Load the dataset and display basic information (shape, columns, data types).
- Identify and report all missing values and duplicate rows.
- Analyze the top 10 best-selling products by quantity and revenue.
- Analyze sales performance by country.
- Plot revenue over time (monthly trend).
- Create a correlation heatmap.
- Detect and visualize outliers using box plots.
- Write at least 5 business insights derived from the data.
- Do not remove any data at this stage – this task is observation only.
- All plots must include proper titles, axis labels, and legends.
- The notebook must run from top to bottom without errors.

Deliverables

- **GitHub Repository Link** – containing all code, documentation, and the data used for the task.
- **LinkedIn Post Link** – containing an explanation of the project and the task.

Evaluation Criteria

Documentation – 10 Marks

- Submit clear step-by-step documentation of the task.
- Include screenshots, implementation details, and explanations where required.

LinkedIn Submission – 50 Marks

- Upload the completed task on LinkedIn.
- Share a professional post explaining the project and internship task.
- Submit the LinkedIn post link through the Google Form provided by the institute.

GitHub Repository / Live Deployment – 10 Marks

- Upload the project to GitHub with proper documentation.
- For applicable domains, deploy the project and provide a live link.
- Submit the GitHub repository link or live project link through the Google Form.

Interview Evaluation – 30 Marks

- An interview will be conducted on Google Meet.
- Students will be evaluated on their understanding, implementation, and problem-solving abilities related to the assigned task.

Total Marks: 100